Proceedings of the International Conference on Cybersecurity & Engineering · Published 2024

Automated Software Vulnerability Discovery via Neural-Guided Symbolic Execution and Fuzzing

Fatima Al-Zahra, Dr. Kimberly Adams, Dr. Maximillian Brandt

Department of Computer Science & Intelligent Systems · Publication Year: 2024

ABSTRACT

Fuzz testing frequently stalls on complex magic-byte comparisons and deep nested conditional branches in compiled binary programs. We present NeuroFuzz-Sym, a hybrid testing pipeline integrating neural branch prediction models with selective concolic symbolic execution. A transformer model trained on disassembled control flow graphs predicts high-yield input seed mutations, invoking the symbolic solver only when fuzzers encounter path stalls. Benchmarked against Google OSS-Fuzz targets, our system discovered 27 zero-day vulnerabilities in open-source libraries with a 4.2x faster branch coverage rate. However, state explosion in the symbolic solver on floating-point arithmetic and cryptographic hash routines remains an unresolved performance barrier.

Keywords— Cybersecurity, Fuzzing, Symbolic Execution, Vulnerability Discovery

INTRODUCTION

Coverage-guided fuzzers generate millions of test inputs per second but struggle to penetrate deep conditional gates. Symbolic execution solves mathematical branch conditions but suffers from catastrophic path explosion.

METHODOLOGY

We employ an attention-based neural predictor to prioritize seed inputs likely to uncover new basic blocks. When branch coverage plateaus, targeted concolic execution solves the immediate bottleneck constraint before handing control back to the fuzzer.

RESULTS

Across the LAVA-M and CGC synthetic benchmarks, NeuroFuzz-Sym achieved 94% branch coverage in 4 hours and uncovered 27 previously unknown memory corruption vulnerabilities (buffer overflows, use-after-free) in live open-source C/C++ projects.

LIMITATIONS

Cryptographic hash operations (e.g., SHA-256 checks) and non-linear floating-point math cannot be inverted by SMT solvers, causing the concolic branch to hang unless explicitly stubbed out.

REFERENCES

[1] Dr. Maximillian Brandt, Lucas Vance, Fatima Al-Zahra (2021). "Adversarial Robustness and Gradient Masking Defenses in Deep Neural Network Classifiers". In Proc. Int. Conf. on Advanced Computing Systems.

[2] Fatima Al-Zahra, Dr. Maximillian Brandt, Chen Wei (2022). "Zero-Knowledge Proofs for Verifiable and Privacy-Preserving SQL Queries in Cloud Databases". In Proc. Int. Conf. on Advanced Computing Systems.

[3] Vaswani, A., et al. (2017). "Attention Is All You Need". Advances in Neural Information Processing Systems, 30.

[4] Kipf, T. N., & Welling, M. (2016). "Semi-Supervised Classification with Graph Convolutional Networks". arXiv:1609.02907.

[5] Lewis, P., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks". NeurIPS.